

Math 3680: Week 5

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1 Equations

I have taken this problem from a set of linear algebra problems I previously wrote for my textbook development position. However, previously, I did not have it in an equation environment. I may go back and standardize that!

Example: Find a linear combination of $\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$ that gives the vector $\begin{bmatrix} 17 \\ -4 \\ 2 \end{bmatrix}$.

Solution: We are looking for a_1 and a_2 such that:

$$a_1 \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ -4 \\ 2 \end{bmatrix}$$

Looking at each dimension separately, we get the system of equations:

$$\begin{aligned} 2a_1 + a_2 &= 17 \\ a_1 - 2a_2 &= -4 \\ -3a_1 + 4a_2 &= 2 \end{aligned}$$

If we can solve this system of equations, we will find a_1 and a_2 . Let's multiply the first equation by 2 and add it to the second equation:

$$\begin{aligned} 2(2a_1 + a_2) + (a_1 - 2a_2) &= 2(17) + (-4) \\ 4a_1 + 2a_2 + a_1 - 2a_2 &= 34 - 4 \\ 5a_1 &= 30 \\ a_1 &= 6 \end{aligned}$$

Now we can take a_1 and substitute it back into any equation in our system to find a_2 . Let's use the third equation:

$$\begin{aligned} -3(6) + 4a_2 &= 2 \\ -18 + 4a_2 &= 2 \\ 4a_2 &= 20 \\ a_2 &= 5 \end{aligned}$$

2 Tables

I had claude make a shortcut for me to create n-sized blocks in my table. A future improvement would be to reduce the amount of `cline` commands needed to make it a more readable and compact table.

Time	Mon	Tue	Wed	Thu	Fri	
8:00	CSC 4880 AI	CSC 3300 PL	CSC 4880 AI	CSC 3300 PL	CSC 4880 AI	
8:30						
9:00						
9:30						
10:00						
10:30	MATH 3680 L ^A T _E X	Redwire (Work)		Redwire (Work)		
11:00						
11:30						
12:00						
12:30						
1:00						
1:30	CSC 4160 SE Capstone		CSC 4160 SE Capstone		CSC 4160 SE Capstone	
2:00						
2:30						
3:00						
3:30						
4:00						
4:30				Front Porch Volunteering!!		
5:00						
5:30	CSC 3665 Databases		CSC 3665 Databases		CSC 3665 Databases	
6:00						
6:30						
7:00						
7:30						
8:00						

Table 1: Weekly Schedule